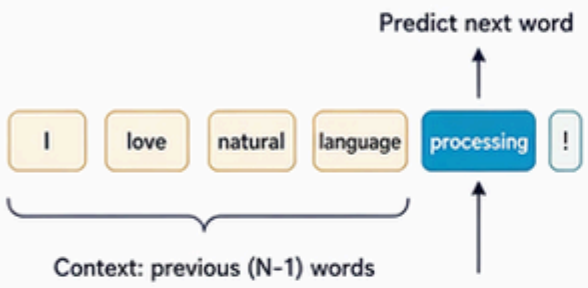
N-GRAM LANGUAGE MODELS

The Building Blocks of Text Prediction

1 CONCEPT

- N-gram models predict a word based on the previous (N-1) words.
- They use the probability of word sequences (n-grams) learned from a large text corpus.
- Simple, efficient, and effective for many NLP tasks.



Examples

Unigram (1-gram): $P(w)$

Bigram (2-gram): $P(w_n \mid w_{n-1})$

Trigram (3-gram): $P(w_n \mid w_{n-2}, w_{n-1})$

2 TRAINING PROCESS

- Collect a large text corpus.
- Tokenize the text into words.
- Extract all n-grams from the corpus.
- Count the frequency of each n-gram and the (n-1)-gram history.
- Calculate probabilities using Maximum Likelihood Estimation (MLE).

Training Workflow

1. **Corpus Collection**
   Gather large text data.
2. **Preprocessing**
   Tokenize, clean the text.
3. **N-gram Extraction**
   Create all n-grams (e.g., bi-grams, tri-grams).
4. **Counting**
   Count frequencies of n-grams and (n-1)-grams.
5. **Probability Calculation**
   $P(w_n \mid \text{history}) = \text{Count}(\text{history}, w_n) / \text{Count}(\text{history})$

3 HOW IT WORKS (EXAMPLE)

Given sentence: *I love natural language processing*

Bigram (2-gram) Example

History (w_{n-1})	Next Word (w_n)	Count	$P(w_n \mid w_{n-1})$
I	love	120	1.00
love	natural	80	0.80
love	machine	20	0.20
natural	language	90	0.90
natural	processing	10	0.10
...	...	...	...

Prediction Example:

If the context is "I", the model predicts "love" with probability 1.00

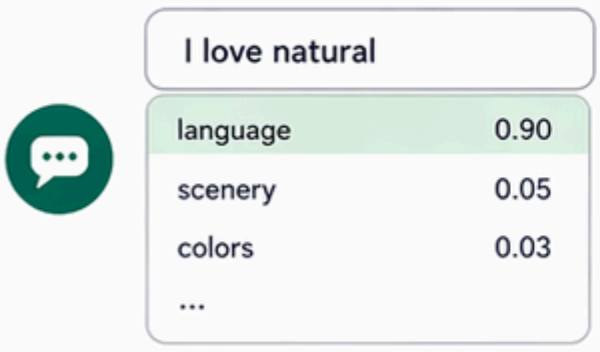
N-gram Model Diagram (Bigram)

Example: $P(\text{natural} \mid \text{love})$

4 TYPICAL APPLICATIONS

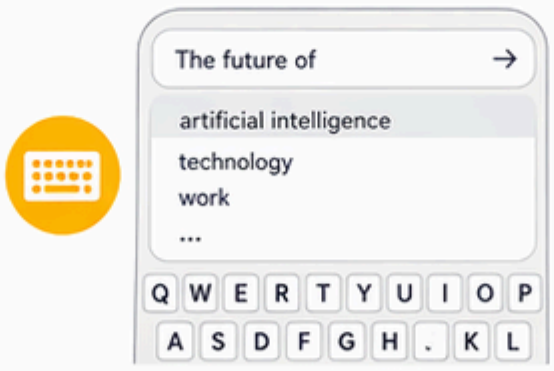
1. Text Prediction

- Predict the next word in a sequence.
- Useful in writing assistants and chatbots.



2. Autocomplete Systems

- Suggest likely next words or phrases.
- Improves typing speed and user experience.



3. Other Applications

Machine Translation
Helps in generating probable translations.

Spell Checking
Detects and suggests corrections.

Language Modeling
Used in speech recognition, text generation, and more.

KEY BENEFITS

- ✔ Simple and easy to implement
- ✔ Computationally efficient
- ✔ Works well with large datasets
- ✔ Foundation for many advanced NLP models

LIMITATIONS

- Limited context (fixed window size)
- Data sparsity for large n
- Does not understand semantics deeply
- Suffers from zero-probability problem

REFERENCES (SCAN QR CODE)

Name : Ishan Vijay Pote | UID : 23011002 | Department : CSE (Data Science)